
ASSESSMENT OF CAVABILITY AND ESTIMATION OF SUPPORT RESISTANCE FOR No. 1 SEAM OF ADRIYALA PROJECT, SCCL

1.0 INTRODUCTION

The SCCL has planned a Longwall Project at Adriyala mine in Godavarikhani, which is at a distance of 65 KM from Karimnagar and 225 KM from Hyderabad. This Project is planned to produce about 2.4 Million tones Per annum (MTPA) with high capacity longwall technology.

The SCCL management has entrusted the work to CIMFR, Dhanbad for, scientific study for assessment of support capacity for the longwall panel of 250m face length at Adriyala Project, SCCL.

The present report deals with the study on the assessment of cavability and estimation of support resistance for working longwall panel of 250m face length at Adriyala Project.

The cavability assessment of strata overlying the No.1 seam of Adriyala Project has been done by using the descriptive borehole litholog and the physico-mechanical properties of the boreholes supplied by the SCCL management.

The caving index of the strongest bed lying above No.1 seam has been used to determine the support resistance requirement for different heights of extraction. Based on the support resistance, the support capacity can be calculated for different support dimension.

2.0 ORIGIN OF THE PROJECT

General Manager, R&D, SCCL has referred the project vide letter no. CRP/RND/B-288/605 dated 30/07/2008.

3.0 OBJECTIVES OF THE PROJECT

The main objectives of the project are: -

- To assess the cavability of the immediate roof rocks over No. 1 seam at Adriyala Project
- Estimation of the support resistance for 250m face length for effective strata control at the face.

4.0 DETAILS OF THE PANEL

Geo-Mining Parameters

Name of the Seam	: No. 1 Seam
Depth of the Seam	: 360 m - 543 m
Thickness of the Seam	: 3.25 m to 6.50 m
Height of Extraction (proposed)	: 3.5 m
Immediate roof	: Coal/Shale overlain by sandstone
Immediate floor	: Sandstone
Existence of overlying worked out area	: Virgin seam II

5.0 ESTIMATION OF CAVABILITY OF ROOF ROCKS

The estimation of cavability of overlying roof rocks is the first step for estimating the requirement of support resistance. There are various norms prevalent in different countries for assessing caving characteristics of roof rocks. By and large, all these norms were scrutinized by CIMFR and were found to be not applicable under Indian situation. The Longwall & Shortwall Mining Division of CIMFR has developed its own method for assessing the cavability of roof rocks. As has been observed, the caving behavior of different beds of overlying roof depends on the thickness, strength and massiveness of the bed. The following empirical relationship expresses the relation between caving index no. 'I' with different factors mentioned above: -

$$I = \frac{\sigma_c L^n t^{0.5}}{5} \dots\dots(1)$$

Where σ_c = Compressive Strength, Kg/cm²

L = Average length of core, cm

t = Thickness of bed, m

n = Constant depending upon the RQD of the bed.

The roof categorization on the basis of cavability index has been given in Table-1.

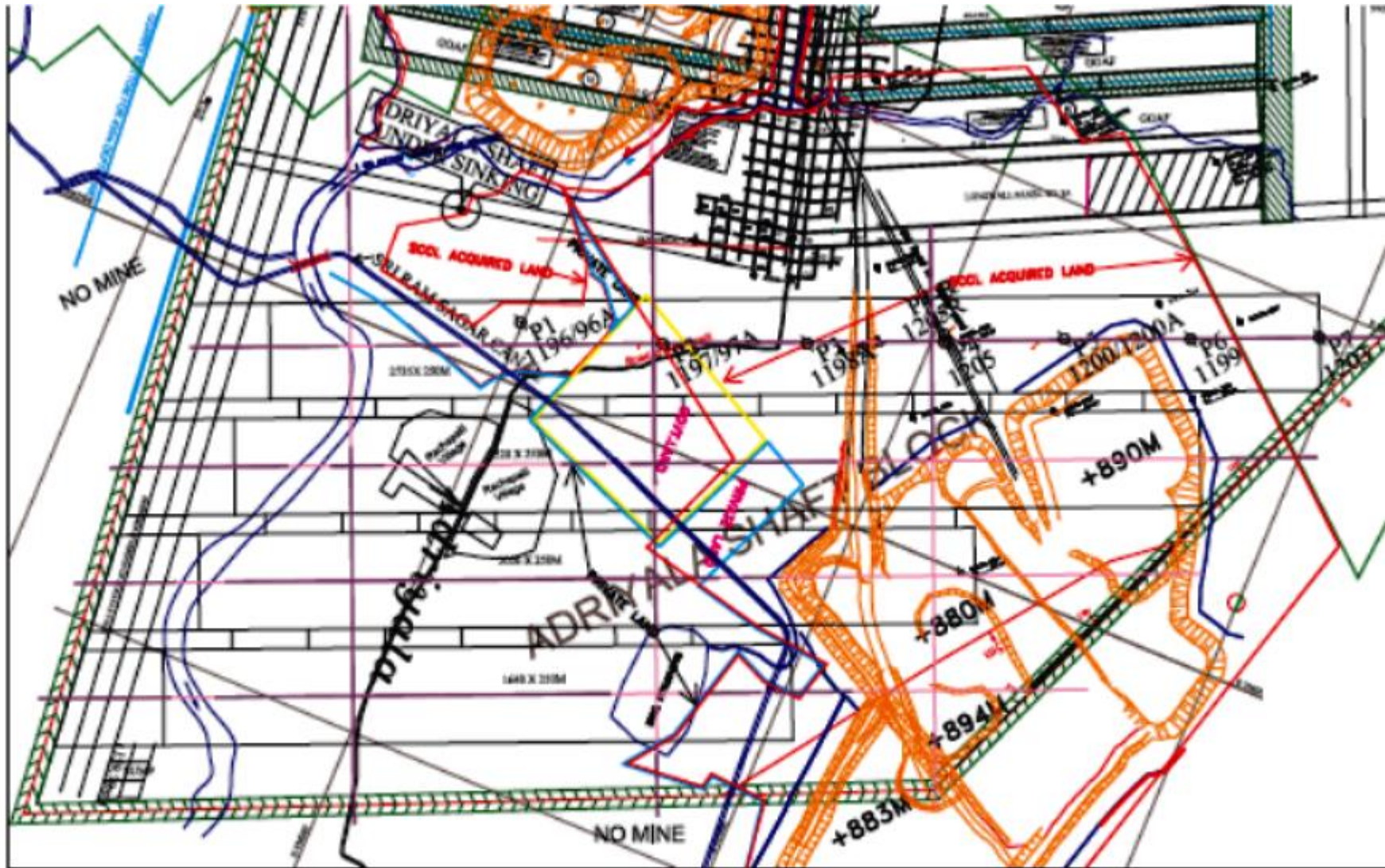


Figure.1a: Plan showing longwall panels in No. 1 Seam of Adriyala block and location of boreholes over the panel.

Table-1: Roof Classification based on Cavability Index

Category	Nature of Caving	Range of Cavability Index	Main fall span, S_m , m	Weighting Dynamism
Category-I	Easily cavable	$I < 2000$	$S_m < 35$	---
Category-II	Moderately cavable	$2000 \leq I < 5000$	$35 \leq S_m < 55$	---
Category-III	Cavable with difficulty	$5000 \leq I < 10000$	$55 \leq S_m < 80$	May/may not be
Category-IV	Cavable with substantial difficulty	$10000 \leq I < 14000$	$80 \leq S_m < 95$	May/may not be
Category-V	Cavable with extreme difficulty	≥ 14000	$S_m \geq 95$	Dynamic

6.0 ESTIMATION OF SUPPORT RESISTANCE

The investigation conducted by CIMFR at caved longwall faces indicates that roof control during normal periods does not pose problem but first weight and periodic weights demand utmost consideration. The roof condition during first weight and periodic weights depends mainly on the following factors:

- (a) Caving properties of the strong bed of sandstone or shale:

This is indicated by 'Caving Index Number' 'I' of the strong bed. This index depends on the thickness of strong beds, its strength properties and massiveness. Higher the index, larger is the span at which first weight (and also subsequent periodic weights) occurs and more severe is the weighting and worse is the resultant roof conditions. Caving index number of all the beds occurring within six to eight times the height of extraction would have to be analyzed to identify the bed, which will have predominant influence in causing the weighting.

- (b) Thickness of cavable beds in between strong bed and the coal seams in terms of height of extraction:

If the thickness of cavable bed is adequate to fill the goaf completely, then a part of the strain energy released during breakage of the strong bed would be absorbed by the caved material in the goaf and severity of first weight and periodic weights would be lessened. But if the thickness of cavable bed is inadequate to fill the goaf then the first weight and the periodic weights would be dynamic and severe in nature unless artificial means of caving is practiced.

- (c) Resistance offered by the support system:

The increase in support resistance has a tendency to decrease the severity of first weight and periodic weights improving the resulting strata control condition.

The influence of three factors mentioned above were separately analyzed from theoretical as well as from observational data and lastly on the basis of multiple regression analysis of the data collected from various longwall faces, it was possible to project the maximum deterioration in the roof condition as a function of the three factors mentioned above. A relationship was evolved correlating maximum deterioration in roof condition (represented by maximum convergence) during first weight and periodic weights with the factors mentioned above. The relationship initially established has been modified slightly with the collection of more data. The relationship is of the following form: [1,2,3]

$$C_m = \left(\frac{A}{P} + 9.6h_e \right) + \left(\frac{kI_{\max}}{k_1 + 1.5} - 23 \right) \quad (2)$$

Where A = Constant depending upon the category of the roof

C_m = Maximum convergence during first weight & periodic weights, mm/m

P = Mean Support resistance, t/m²

$I_{\max}$ = Cavability Index of strong bed,

K_1 = A factor depending on the thickness 't' of cavable bed in between strong bed and the coal seam in terms of height of extraction.

h_e = height of extraction, m

The above relationship has two components. The first component of hyperbolic nature is a function of support and height of extraction and expresses the influence of support system and height of extraction on convergence whereas the second component expresses the influence of nature, properties & disposition of strata at the face.

On careful scrutiny, it would be observed that at a particular mining condition, C_m is related to support resistance by a hyperbolic curve but the point at which steep rise in convergence would occur or convergence attains an asymptotic trend would vary from one situation to another.

With higher value of 't' the value of K_1 is increased (the degree of filling in the goaf is improved) and the value of ' C_m ' is reduced. One way to increase the value of 't' is to

resort to blasting of immediate roof in the goaf. The amount of reduction in support density may be calculated on the basis of suggested relationship.

With the help of the suggested relationship, it is possible to project maximum convergence at a proposed longwall face (i.e. under a particular strata condition) with different values of mean load density. As the rates of convergence have already been correlated with degradation of roof on the basis of measurements conducted over a few hundreds of cycles it is possible to project likely roof condition with a particular value of mean load density. A value of mean load density would be chosen which would give safe roof condition during the occasions of weighting. The value of convergence vis-a-vis roof condition has been given below:

Maximum Convergence	Expected roof condition
Maximum convergence 60 mm/m	Convergence within permissible limit
Above 60mm/m to 100mm/m	Minor roof fracturing increasing with the value of convergence.
Above 100 up to 160 mm/m	Significant roof fracturing and roof degradation. Seriousness increasing with increase in convergence.
Above 160 mm/m	Rock fall zone.

For Indian geo-mining conditions as shown in the above table, for limited and acceptable degradation of the roof a maximum convergence at the face is taken as 60 mm/m. Effective Support Resistance for different height of extraction is calculated based on the above relationship [2].

The empirical relation described above for estimation of support capacity has the following draw backs:

- The model is discontinuous because of the factor A and K_1
- The model was found very difficult to use by practicing mining engineers
- The formation of the beds and its methodology were not well defined

In view of the above draw backs, in an joint S&T project undertaken by CIMFR Dhanbad, ISM Dhanbad and NIRM Kolar, funded by Ministry of Coal, Govt. of India, parting plane approach for identification of beds and caving layers and modified

empirical relations have been developed for estimating the maximum convergence slope [Eq. 4] during first weight & periodic weights in mm/m and rated support density [Eq. 5] required for longwall face.

Rated support resistance

However the Rated Support Resistance should take into account the following deficiencies during the actual operation –

I.	Leakage in leg circuit	:	10%
II.	Setting load deficiencies	:	10%
III.	Miscellaneous (deviation from normal span, Premature bleeding of leg circuits, etc.)	:	10%

Hence the overall deficiency to be taken into account is 30% and thus, the Rated Support Resistance is calculated from the Mean Support Resistance with a safety factor of 1.3.

7.0 PHYSICO-MECHANICAL PROPERTIES OF THE OVERLYING ROOF ROCKS

For the cavability assessment of the overlying roof rocks of No.1 Seam and estimation of support resistance requirement at Adriyala Block, geo-technical data from six boreholes namely 1203, 1199,1205R,1198A,1196A, and 1210 are considered. For the cavability assessment of the overlying roof rocks of No.1 Seam and estimation of support resistance requirement at No.1 Seam of Adriyala Block, geo-technical data from the above boreholes are considered.

The lithology of different boreholes have been analysed to get the major Parting Planes overlying the seam and to estimate the Immediate roof and main roof in the different boreholes so as to get an idea of variation in strata over the coal seam. The immediate roof is the fractured lower roof, which readily caves behind the supports forming the goaf. During the fracturing process, the main roof can induce vertical load on the supports and periodic falls. Table 2 summarizes the findings. The details of these six boreholes are given below.

The physico-mechanical properties of the different rock types overlying No. 1 seam of Adriyala block, as determined at SCCL laboratory, have been used in the study. Based on the compilation of test data supplied by the mine for Borehole No. 1203, 1199, 1205R, 1198A, 1196A and 1210 it is observed that the test data for different rock types shows a very wide variation. In view of this, only those data are considered for the design purpose which varies within 25% of the maximum value for a given rock type under

consideration. With this consideration, the average strength properties have been calculated from combining the data of all the boreholes mentioned above. The average rock properties thus obtained are given in Table 3

Table-2: Variation in depth of cover, seam thickness and location of coal band (major weakness plane) above No. 1 seam

Sl. No	Bore hole No.	Location	Depth (m)	Seam thickness (m)	Thickness of Immediate roof, m	Thickness of Main roof, (m)	Parting Planes above Seam 1, (m)
1.	1203	At the start along the central line of 1 st panel.	360.88	5.88	4.3	14.3	21 m, 28 m, 48m
2.	1199	Approx. 100 m along the central line from the start of 1 st panel.	374.91	5.92	2.42	21.36	24 m, 28 m, 32m, 48m
3.	1205R	Along the central line and at the middle of 1 st panel.	390.97	6.48	6.95	24.33	25 m, 32 m, 47 m
4.	1198A	Along the central line and at the middle of 1 st panel.	406.72	5.68	3.51	20.42	24 m, 31 m, 46m
5.	1196A	At the end of 1 st panel along the central line.	421.68	6.31	12.49	17.0	34 m, 36 m, 41 m
6.	1210	Near the mine boundary at the start of 4 th panel in Adriyala block.	542.80	6.12	3.59	26.49	30 m

The physico-mechanical properties of the different rock beds for the six boreholes 1203, 1199, 1205R, 1198A, 1196A and 1210 are given in table 3(a) to 3(f) respectively. The histograms as shown in Figure. 3(a) to 3(f) depict the borehole sections and the variation in compressive strength and RQD for the boreholes. The thickness of the No. 1 seam varies between 5.88m to 6.48m in the 6 boreholes.

With a height of extraction of 3.5m, a considerable thickness more than 2.5m of inferior coal will form the immediate roof.

The first major parting plane at the start of the first panel in Adriyala lies 21m above the coal seam in BH 1203. This gradually rises above the seam to 34.03m in BH 1196.

Table-3: Average physico-mechanical properties of different rock types of Adriyala Mine, SCC

Lithology	Comp.	Tensile	Density,
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	strength, Kg/cm ²	strength Kg/cm ²	gm/cc
cvcgsst_pebbly	98.00	6.53	2030
vcsst	112.48	7.50	2030
cvcgsst	126.96	8.46	2030
cgsst	146.17	10.30	2030
mcsst	185.00	27.00	2073
mgsst	240.33	33.90	2035
fmgssst	287.00	38.80	2060
fgsst	312.00	43.80	2240
int_sh_sst	269.00		
sh	40.00		
carb_sh	40.00		
shcoal	300.00		
coal	281.00		
carb_clay	94.00		

8.0 CAVABILITY ASSESSMENT

Borehole No – 1203

The 1st major parting bed of carbonaceous clay (Bed no. 5) which is of thickness 2.47 m and at a height of 21, m above the coal seam.

Overlying the first parting plane the 2nd parting plane of thickness 3.79 m consisting of coal, shale and carbonaceous clay is lying at a height of 28, m above the coal seam.

The 1st and 2nd parting planes are separated by a sandstone bed of 4.25 m. In between the coal seam and the 1st major parting plane there are one competent sandstone bed of thickness 14.34 m, with a maximum Caving Index of 4192.

Borehole No – 1199

Similar to BH No. 1203 the 1st major parting bed of thickness 0.73, m consisting of coal and carbonaceous sandstone which lies at a height of 24.8, m above the coal seam.

The 2nd parting plane of coal, shale and clay thickness 6.43 m, lies at a height of 28.3 m above the coal seam. The two parting planes are separated by a sandstone bed of 2.77, m. Similar to the previous borehole above the coal seam and the 1st major parting plane there is a competent sandstone bed of thickness 21.36 m with a maximum Caving Index of 4918.

Borehole No – 1205R

Similar to previous two boreholes the 1st major parting plane of thickness 5.38 m consisting of coal, shale and clay which lies at a height of 24.3 m above the coal seam.

The 2nd parting plane of thickness 3.33 m lies at a height of 30.95 m. The two parting planes are separated by a sandstone bed of thickness 1.24 m.

Overlying the coal seam, similar to the previous boreholes there are three competent sandstone beds of thickness 2.14m, 9.47m and 7.29m with a Caving Index of 1958, 4629 and 4021 respectively.

Borehole No – 1198A

Similar to previous boreholes the 1st major parting plane of thickness 3.86, m consisting of coal and shale which lies at a height of 24 m above the coal seam.

The 2nd parting plane of thickness 1.81 m lies at a height of 31 m above the coal seam. These two parting planes are separated by a sandstone bed of thickness 3.12 m.

Overlying the coal seam and the 1st major parting plane there are three competent sandstone bed of thickness 7.89 m, 2.93 m and 6.0 m separated by a weak sandstone bed of thickness 3.07 m with a maximum Caving Index of 2792.

Borehole No – 1196A

The 1st major parting plane which is present in the previous boreholes has moved up at a height of 34 m above the coal seam and a thin parting of 0.54 m consisting of Shale.

The 2nd parting plane of thickness 3.79 m of carbonaceous coal and clay lies at a height of 35.6 m above the coal seam. These two parting beds are separated by a thin sandstone bed of thickness 1.08 m.

Overlying the coal seam and below the 1st major parting plane there are four competent sandstone bed of thickness 2.79 m, 2.95 m, 17.0 m, and 2.74 m. These beds are separated by weak sandstone beds of thickness 3.05 m and 1.8 m of low RQD. The maximum Caving Index observed within the caving zone is 6326.

Borehole No – 1210

The 1st major parting plane as observed in previous boreholes has merged with 2nd parting plane with a total thickness of 6.38 m consisting of coal, shale and clay lying at a height of 30 m above the coal seam.

Overlying the coal seam and below the 1st major parting plane there are four competent sandstone bed of thickness 7.27 m, 3.33 m, 9.0 m, and 6.17 m with caving index of 4907, 3132, 3801 and 6294. The maximum Caving Index within the caving zone is 6294.

The maximum Caving Index Number (I_{max}) of 6326 is observed for the beds within the caving zone overlying No. 1 seam in Adriyala Block. Based on I_{max} , caving index of the strongest bed, the immediate roof above No. 1 seam is classified as “*cavable with difficulty*”. The maximum thickness of the main roof in the first panel is 24.3m which consists of coarse to medium grained sand stone. The RQD of the competent bed in the main roof is more than 90 and maximum thickness of the competent bed is 17m comprising of mostly medium to coarse grained sand stone with average compressive strength of 192 kg/cm².

Table 3(a1): Physico-mechanical properties of different rock types as per B.H.No.1203 of Adriyala block, GDK 10A Mine, SCCL

Bed No	Rock Type	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
		from	to	from	to								
	coal	360.88	364.38	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
1	shcoal	356.61	360.88	0.00	4.27	4.27	82	18.3	--	232	--	1752	43
2	cvcgsst	342.27	356.61	4.27	18.61	14.34	92	21.49	2030	132	9	4192	56
3	cgsst	340.44	342.27	18.61	20.44	1.83	59	10.49	2030	146	10	621	36
4	cgsst	339.61	340.44	20.44	21.27	0.83	87	19.88	2030	146	10	962	50
5	carb_clay	337.14	339.61	21.27	23.74	2.47	84	18.83	--	94	--	556	42
6	cgsst	332.89	337.14	23.74	27.99	4.25	91	21.22	2031	162	14	3004	58
7	shcoal	329.10	332.89	27.99	31.78	3.79	69	13.9	--	184	--	995	42
8	mgsst	328.12	329.1	31.78	32.76	0.98	98	23.5	2035	240	34	2103	68
9	shcoal	326.27	328.12	32.76	34.61	1.85	68	13.47	--	273	--	999	43
10	cvcgsst	313.11	326.27	34.61	47.77	13.16	87	19.86	2045	130	11	3660	47
11	shcoal	307.64	313.11	47.77	53.24	5.47	81	18.02	--	119	--	1162	43
12	cgsst	305.60	307.64	53.24	55.28	2.04	63	11.86	2030	146	10	1189	39
13	cgsst	301.12	305.60	55.28	59.76	4.48	91	21.3	2030	146	10	2430	57

Table 3(a2): Caving Layers as per B.H.No.1203 of Adriyala block GDK 10A Incline Mine, SCCL

Bed No	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
	from	to	from	to								
No.1 Seam	360.88	364.38	-3.50	0.00	3.5	40	4.37	--	281	--	0	43
Immediate roof	356.61	360.88	0.00	4.27	4.27	82	18.3	--	232	--	1752	43
Main roof	342.27	356.61	4.27	18.61	14.34	92	21.49	2030	132	9	4192	56
Upper main roof	340.44	313.11	18.61	53.24	34.63	82	18.28	2039	148	13	2261	47
Caving Layer 4	305.6	305.6	53.24	59.76	6.52	82	18.35	2030	146	10	2042	52

Table 3(b1) : Physico-mechanical properties of different rock types as per B.H.No.1199 of Adriyala block, GDK 10A Mine, SCCL													
Bed No	Avg Rock Type	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
		from	to	from	to								
	coal	374.91	378.41	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
1	shcoal	372.49	374.91	0.00	2.42	2.42	43	5.49	--	281	--	480	43
2	cvcgsst	351.13	372.49	2.42	23.78	21.36	91	21.13	2033	134	10	4918	55
3	mgsst	350.07	351.13	23.78	24.84	1.06	50	7.66	2136	229	28	601	37
4	carb_clay	349.34	350.07	24.84	25.57	0.73	89	20.56	--	227	--	890	47
5	mgsst	346.57	349.34	25.57	28.34	2.77	89	20.6	2081	214	28	2686	55
6	shcoal	340.14	346.57	28.34	34.77	6.43	85	19.3	--	156	--	1526	44
7	cvcgsst	336.91	340.14	34.77	38.00	3.23	90	20.9	2030	127	8	2028	54
8	cvcgsst	332.97	336.91	38.00	41.94	3.94	53	8.51	2030	127	8	1645	31
9	cgsst	327.23	332.97	41.94	47.68	5.74	88	20.15	2052	163	19	3603	51
10	shcoal	319.9	327.23	47.68	55.01	7.33	93	22.01	--	72	--	859	44
11	cgsst	315.58	319.9	55.01	59.33	4.32	50	7.67	2046	171	20	668	32

Table 3(b2) : Caving Layers as per B.H.No.1199 of Adriyala block GDK 10A Incline Mine, SCCL												
Bed No	Depth		Ht. above coal		Bed	RQD,	Avg.	Density,	CS,	TS,	CI	RMR
	from surface, m		seam,	thickness,								
	from	to	from	to	m	%	Core Length cm	gm/cc	ksc	ksc		
No.1 Seam	374.91	378.41	-3.50	0.00	3.5	40	4.37	--	281	--	0	43
Immediate roof	372.49	374.91	0.00	2.42	2.42	43	5.49	--	281	--	480	43
Main roof	351.13	372.49	2.42	23.78	21.36	91	21.13	2033	134	10	4918	55
Upper main roof	350.07	327.23	23.78	55.01	31.23	83	18.65	2053	140	17	1875	45
Caving Layer 4	315.58	319.9	55.01	59.33	4.32	50	7.67	2046	171	20	668	32

Table 3(c1) : Physico-mechanical properties of different rock types as per B.H.No.1205R of Adriyala block, GDK 10A Mine, SCCL

Bed No	Avg Rock Type	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
		from	to	from	to								
	coal	390.97	394.47	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
1	shcoal	387.26	390.97	0.00	3.71	3.71	48	7.10	--	220	--	601	43
2	cgsst	385.12	387.26	3.71	5.85	2.14	91	21.26	2034	171	18	1958	58
3	cvcgsst	384.02	385.12	5.85	6.95	1.10	80	17.54	2030	127	8	829	43
4	mcgsst	374.55	384.02	6.95	16.42	9.47	96	22.71	2064	177	24	4629	63
5	mcgsst	373.93	374.55	16.42	17.04	0.62	73	15.29	2073	185	27	1036	48
6	mcgsst	366.64	373.93	17.04	24.33	7.29	92	21.48	2056	188	25	4021	60
7	coal_sh_clay	361.26	366.64	24.33	29.71	5.38	88	20.31	--	151	--	1424	44
8	fgsst	360.02	361.26	29.71	30.95	1.24	84	18.85	2240	312	44	2357	59
9	carb_sh	356.69	360.02	30.95	34.28	3.33	89	20.41	--	219	--	1635	45
10	mcgsst	344.89	356.69	34.28	46.08	11.80	88	20.26	2075	191	23	5756	52
11	shcoal	336.51	344.89	46.08	54.46	8.38	82	18.19	--	--	--	--	44
12	mcgsst	335.02	336.51	54.46	55.95	1.49	80	17.53	2034	215	28	1632	49

Table 3(c2) : Caving Layers as per B.H.No. 1205R of Adriyala block GDK 10A Incline Mine, SCCL

Bed No	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
	from	to	from	to								
No.1 Seam	390.97	394.47	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
Immediate roof	387.26	385.12	0.00	6.95	6.95	66	13.11	2033	190	15	1055	47
Main roof	374.55	373.93	6.95	24.33	17.38	93	21.93	2061	182	24	4247	61
Upper main roof	361.26	344.89	24.33	54.46	30.13	86	19.65	2091	192	25	3860	48
Caving Layer 4	335.02	336.51	54.46	55.95	1.49	80	17.53	2034	215	28	1632	49

Table 3(d1) : Physico-mechanical properties of different rock types as per B.H.No.1198A of Adriyala block, GDK 10A Mine, SCCL

Bed No	Avg Rock Type	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
		from	to	from	to								
	coal	406.72	410.22	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
1	sh_coal	403.21	406.72	0.00	3.51	3.51	64	12.23	--	162	--	745	43
2	cgsst	395.32	403.21	3.51	11.4	7.89	80	17.52	2049	160	17	2792	45
3	cgsst	392.25	395.32	11.40	14.47	3.07	39	4.18	2040	145	13	0	27
4	cgsst	389.32	392.25	14.47	17.40	2.93	91	21.13	2037	164	15	2188	57
5	vcsst	383.32	389.32	17.40	23.40	6.00	87	19.77	2030	115	8	2021	46
6	fmgssst	382.79	383.32	23.40	23.93	0.53	100	24.17	2060	287	39	1910	73
7	sh_coal	378.93	382.79	23.93	27.79	3.86	92	21.70	--	162	--	1382	43
8	fmgssst	375.81	378.93	27.79	30.91	3.12	91	21.21	2047	263	36	3631	64
9	sandy_sh	374	375.81	30.91	32.72	1.81	90	20.89	--	--	--	--	45
10	cgsst	371.32	374	32.72	35.4	2.68	69	13.95	2040	146	13	1129	42
11	cvcgsst	369.45	371.32	35.40	37.27	1.87	99	24.00	2030	127	8	1573	62
12	mcgsst	368.32	369.45	37.27	38.4	1.13	50	7.53	2068	178	25	1095	34
13	mcgsst	366.82	368.32	38.40	39.90	1.5	94	22.19	2073	185	27	1870	62
14	mcgsst	360.82	366.82	39.90	45.90	6.00	46	6.35	2043	189	21	1445	32
15	carb_clay	357.72	360.82	45.90	49.00	3.10	58	10.33	--	94	--	342	42
16	mcgsst	354.71	357.72	49.00	52.01	3.01	63	11.89	2058	171	21	1579	41

Table 3(d2) : Caving Layers as per B.H.No. 1198A of Adriyala block GDK 10A Incline Mine, SCCL

Bed No	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
	from	to	from	to								
No.1 Seam	406.72	410.22	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
Immediate roof	403.21	406.72	0.00	3.51	3.51	64	12.23	--	162	--	745	43
Main roof	395.32	383.32	3.51	23.93	20.42	78	16.87	2041	148	14	2396	45
Upper main roof	378.93	357.72	23.93	52.01	28.08	72	14.88	2048	171	22	1567	45
Caving Layer 4	--	--	--	--	--	--	--	--	--	--	--	--

Table 3(e1) : Physico-mechanical properties of different rock types as per B.H.No.1196 of Adriyala block, GDK 10A Mine, SCCL

Bed No	Avg Rock Type	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
		from	to	from	to								
	coal	421.68	425.18	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
1	shcoal	417.98	421.68	0.00	3.70	3.70	52	8.3	--	215	--	686	42
2	mcsst	415.19	417.98	3.70	6.49	2.79	82	18.26	2059	209	27	2276	50
3	cgsst	412.24	415.19	6.49	9.44	2.95	88	20.27	2041	163	16	2070	50
4	cvcgsst	409.19	412.24	9.44	12.49	3.05	56	9.56	2031	128	9	948	33
5	mcsst	392.19	409.19	12.49	29.49	17.00	89	20.55	2055	192	24	6326	53
6	mcsst	390.39	392.19	29.49	31.29	1.80	68	13.54	2073	185	27	1593	45
7	mgsst	387.65	390.39	31.29	34.03	2.74	88	20.32	2071	248	34	3041	57
8	shale	387.11	387.65	34.03	34.57	0.54	98	23.56	--	40	--	139	46
9	fgsst	386.03	387.11	34.57	35.65	1.08	92	21.42	2240	312	44	2564	69
10	carb_coal_clay	382.24	386.03	35.65	39.44	3.79	96	22.87	--	173	--	1538	43
11	fmgssst	380.56	382.24	39.44	41.12	1.68	43	5.51	2063	277	38	1293	38
12	sandy_sh	378.47	380.56	41.12	43.21	2.09	99	23.86	--	192	--	1328	45
13	cgsst	373.19	378.47	43.21	48.49	5.28	95	22.61	2036	177	18	3426	62

Table 3(e2) : Caving Layers as per B.H.No. 1196 of Adriyala block GDK 10A Incline Mine, SCCL

Bed No	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
	from	to	from	to								
No.1 Seam	421.68	425.18	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
Immediate roof	417.98	412.24	0.00	12.49	12.49	68	13.66	2043	180	17	1432	44
Main roof	392.19	409.19	12.49	29.49	17.00	89	20.55	2055	192	24	6326	53
Upper main roof	390.39	378.47	29.49	48.49	19.00	88	20.06	2070	201	28	2259	52
Caving Layer 4	--	--	--	--	--	--	--	--	--	--	--	--

Table 3(f1) : Physico-mechanical properties of different rock types as per B.H.No.1210 of Adriyala block, GDK 10A Mine, SCCL

Bed No	Avg Rock Type	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length, cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
		from	to	from	to								
	coal	542.8	546.3	-3.50	0.00	3.50	40	4.37	--	281	--	0	43
1	coal	539.21	542.8	0.00	3.59	3.59	53	8.64	--	197	--	645	43
2	mgsst	531.94	539.21	3.59	10.86	7.27	95	22.53	2053	217	31	4907	65
3	mcsst	531.22	531.94	10.86	11.58	0.72	53	8.67	2073	185	27	345	36
4	mgsst	527.89	531.22	11.58	14.91	3.33	86	19.68	2044	240	34	3132	55
5	mgsst	518.89	527.89	14.91	23.91	9.00	74	15.74	2056	222	31	3801	51
6	fmgst	512.72	518.89	23.91	30.08	6.17	93	21.87	2105	274	38	6294	67
7	shcoal	506.34	512.72	30.08	36.46	6.38	90	20.92	--	182	--	1924	44
8	mcsst	502.14	506.34	36.46	40.66	4.20	87	20.01	2058	204	28	3039	53
9	cvcgst	501.12	502.14	40.66	41.68	1.02	73	15.12	2041	142	13	745	44
10	cgsst	498.39	501.12	41.68	44.41	2.73	97	23.09	2042	151	14	2362	62

Table 3(f2) : Caving Layers as per B.H.No. 1210 of Adriyala block GDK 10A Incline Mine, SCCL

Bed No	Depth from surface, m		Ht. above coal seam, m		Bed thickness, m	RQD, %	Avg. Core Length cm	Density, gm/cc	CS, ksc	TS, ksc	CI	RMR
	from	to	from	to								
No.1 Seam	542.8	546.3	-3.50	0.00	3.5	40	4.37	--	281	--	0	43
Immediate roof	539.21	542.8	0.00	3.59	3.59	53	8.64	--	197	--	645	43
Main roof	531.94	518.89	3.59	30.08	26.49	85	19.34	2065	234	33	4508	58
Upper main roof	506.34	501.12	30.08	44.41	14.33	89	20.65	2050	180	21	2251	50
Caving Layer 4	--	--	--	--	--	--	--	--	--	--	--	--

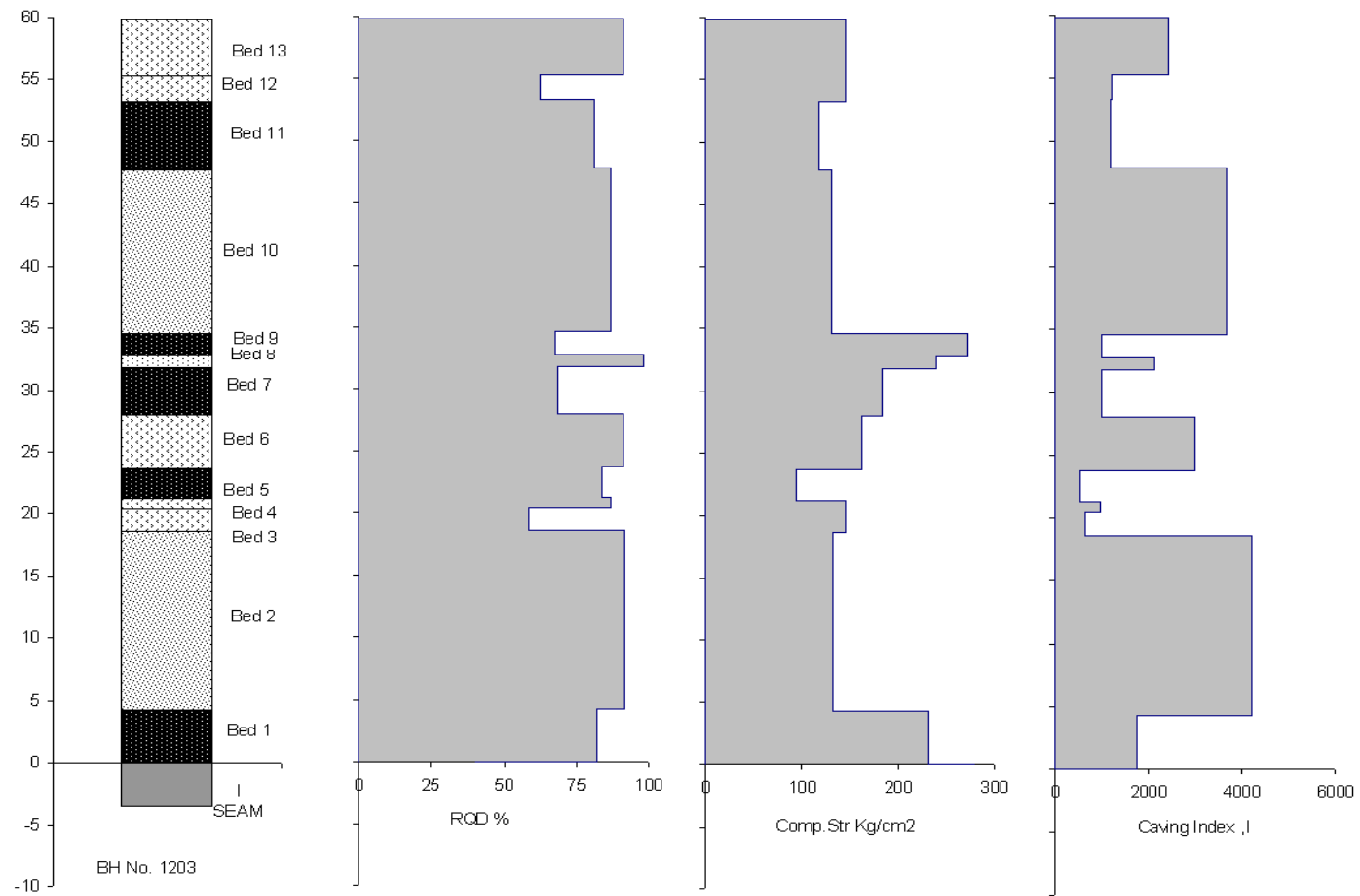


Figure. 3(a): Histogram of physico-mechanical properties and caving index for different rock beds overlying No.1 seam at Adriyala Mines as per borehole no. 1203.

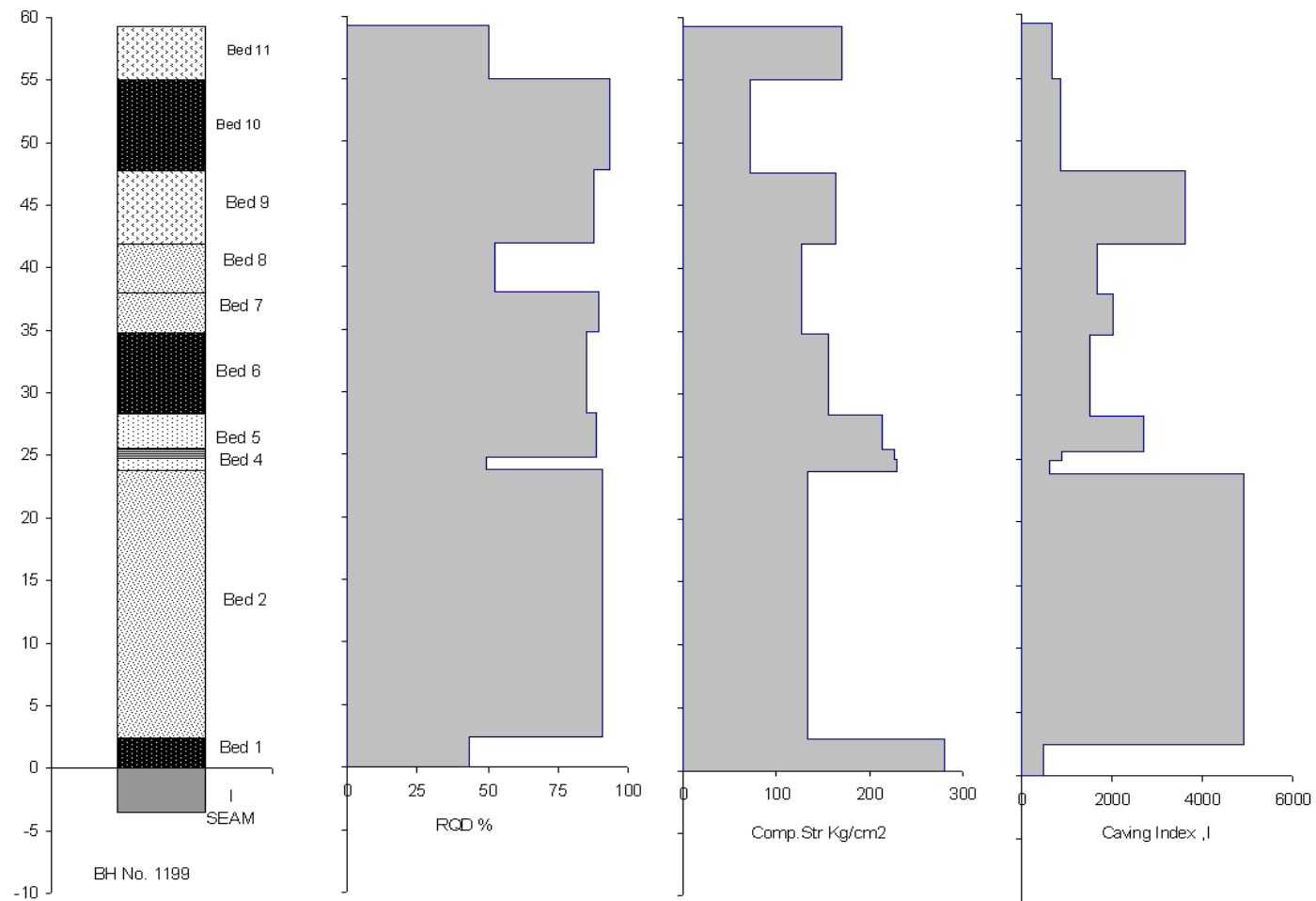


Figure. 3(b): Histogram of physico-mechanical properties and caving index for different rock beds overlying No.1 seam at Adriyala Mines as per borehole 1199

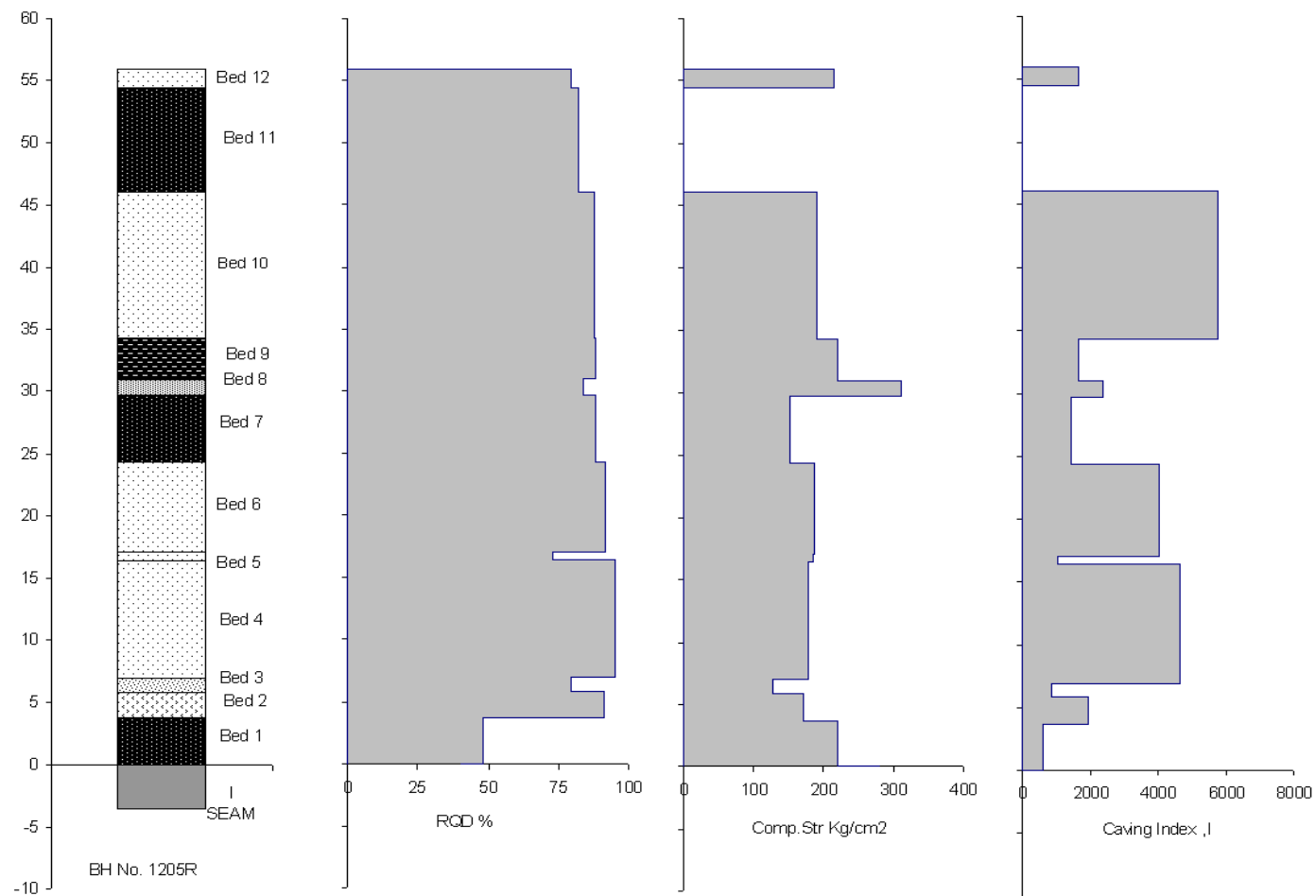


Figure. 3(c): Histogram of physico-mechanical properties and caving index for different rock beds overlying No.1 seam at Adriyala Mines as per borehole no. 1205R .

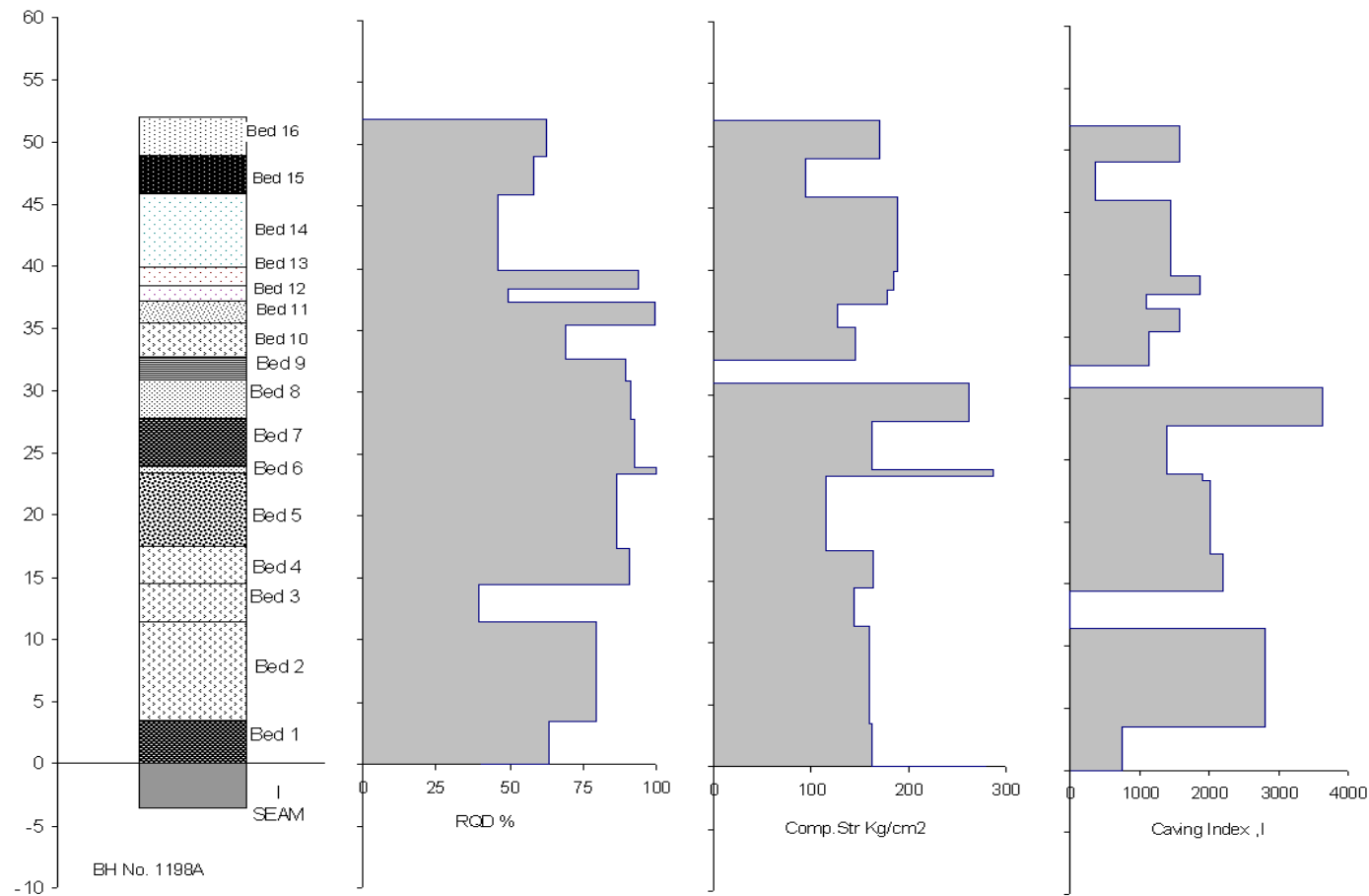


Figure. 3(d): Histogram of physico-mechanical properties and caving index for different rock beds overlying No.1 seam at Adriyala Mines as per borehole no. 1198A.

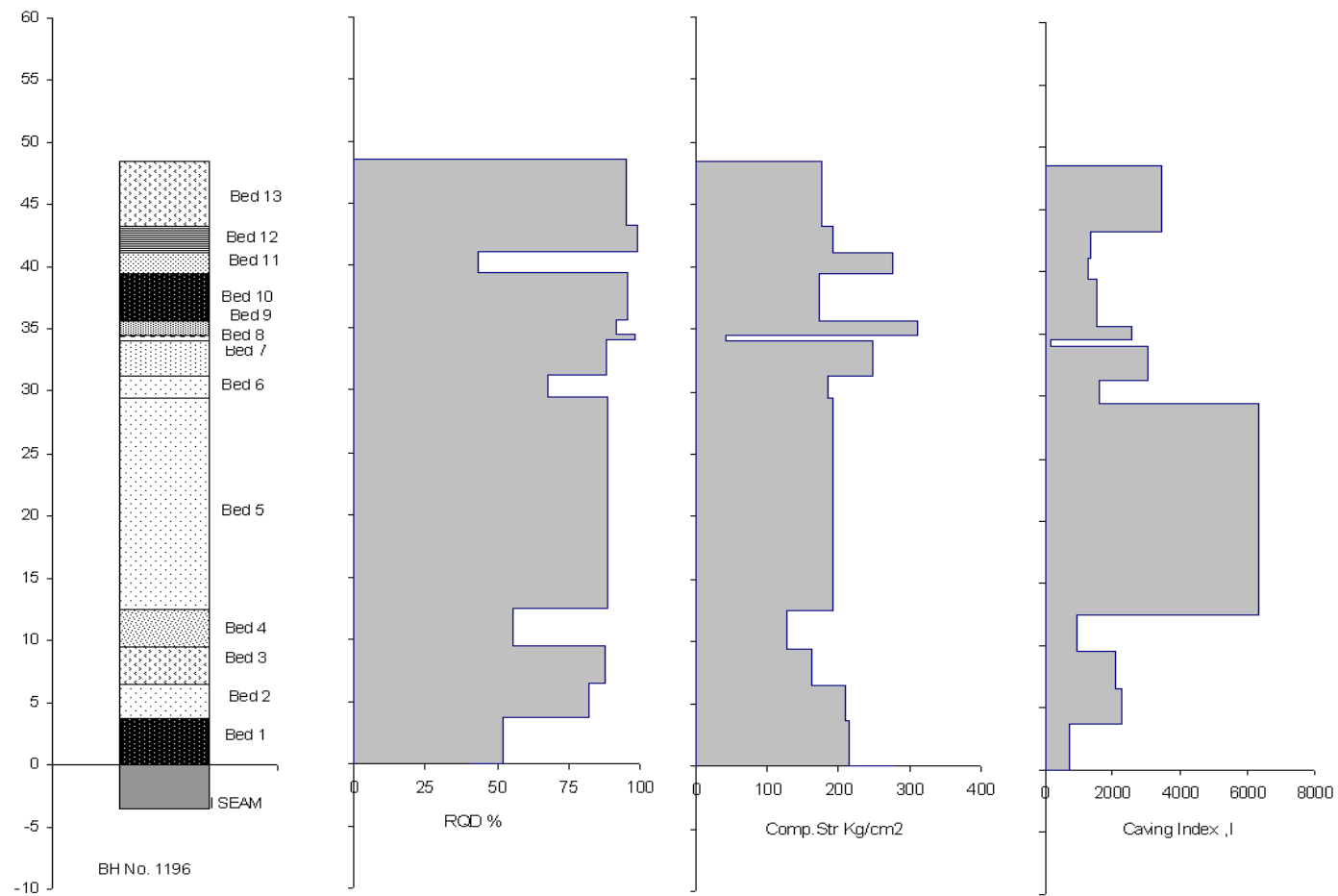


Figure. 3(e): Histogram of physico-mechanical properties and caving index for different rock beds overlying No.1 seam at Adriyala Mines as per borehole no. 1196.

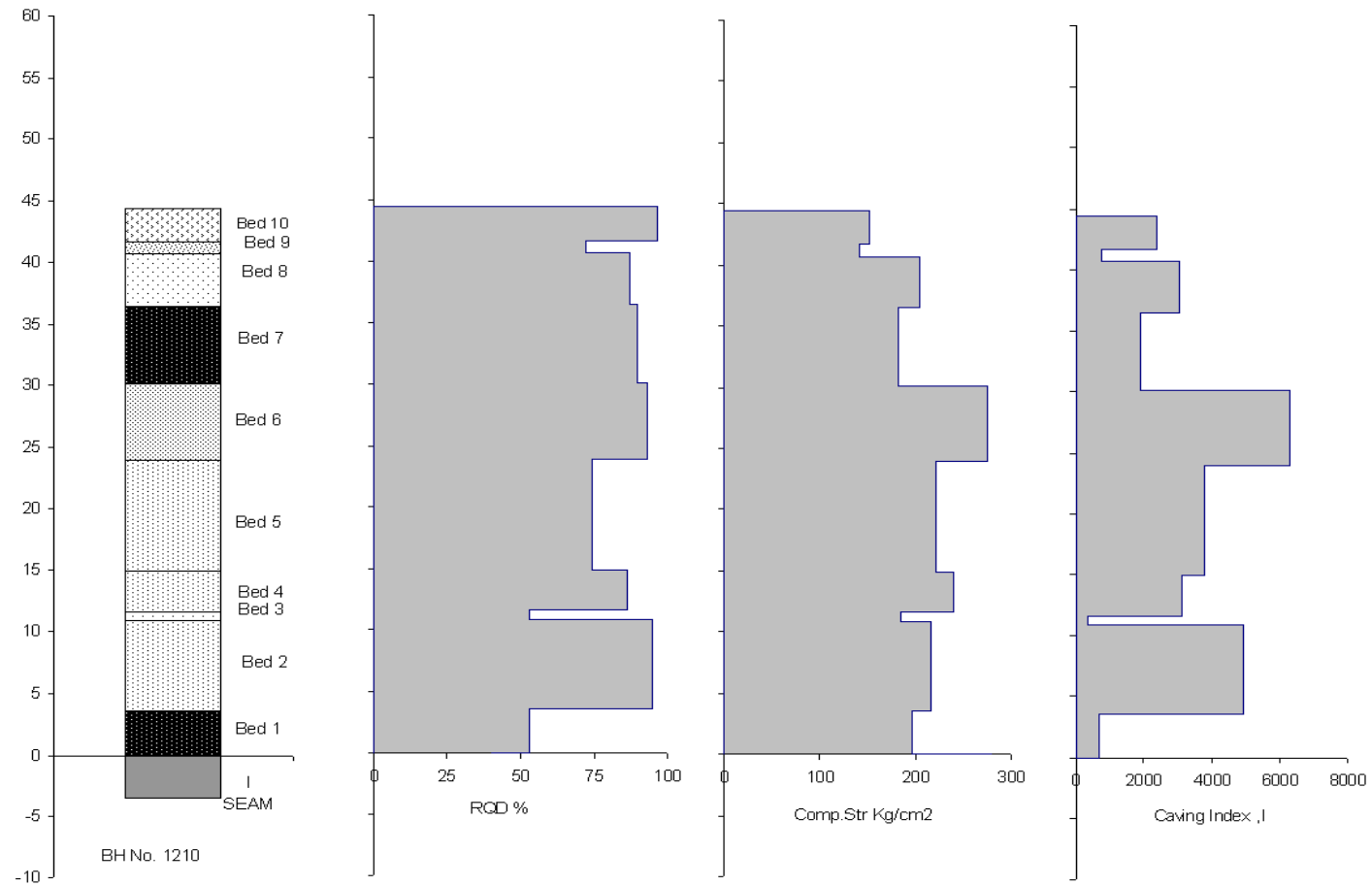


Figure. 3(f): Histogram of physico-mechanical properties and caving index for different rock beds overlying No.1 seam at Adriyala Mines as per borehole no. 1210.

Table- 4: Required Support Resistance for Adriyala Block		
Height of Extraction, (m)	Effective Support Resistance , P_e (T/m ²)	Rated Support Resistance, P_r (T/m ²)
2.8	59	77
2.9	62	81
3	64	83
3.1	66	86
3.2	69	90
3.3	72	94
3.4	75	98
3.5	79	103
3.6	83	107

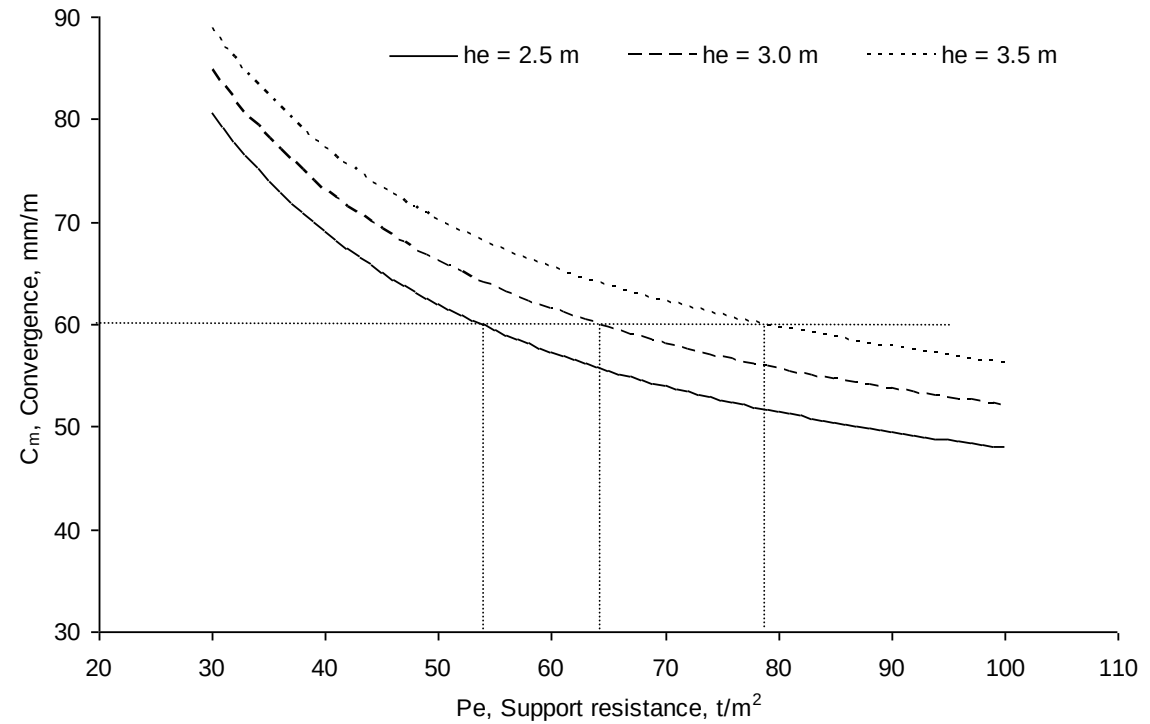


Figure 4: Effective support resistance vs. convergence at the face for Adriyala Block

9.0 SUPPORT RESISTANCE ESTIMATION

The six boreholes No. 1203, 1199, 1205R, 1198A, 1196A and 1210 were studied and the variations of the characteristics of the competent beds within the caving zone are tabulated in Table 5.

Sl no.	Borehole number	No. of competent beds	Rock Type	Variation in RQD	Variation in Compressive strength, kg/cm ²	Variation in Caving Index number	I_{max}
1.	1203	1 (14.3m)	Coarse to very coarse grained sst.	92	132	4192	4192
2.	1199	1 (21.4m)	Coarse to very coarse grained sst.	91	134	4918	4918
3.	1205R	3	Coarse to medium grained sst.	91-96	171-188	1958-4629	4629
4.	1198A	3	Coarse to medium grained sst.	80-91	115-164	2021-2792	2792
5	1196	4	Coarse to medium grained sst	82-89	163-248	2070-6326	6326
6	1210	4	Coarse to medium grained sst	74-95	217-274	3132-6294	6294

The maximum Caving Index Number of 6326 is observed for the beds within the caving zone overlying No. 1 seam in Adriyala Block. The Effective Support Resistance (P_e) and Rated Support Resistance (P_r) required at the face at different height of extraction are calculated using equation 4 and are given in Table-4. The Convergence Vs Effective Support Resistance for different heights of extraction for Eq. 4 is shown in Figure. 4.

The rated support resistance calculated from the Effective Support Resistance with a safety factor of 1.3 for a height of extraction of 3.5m based on maximum convergence of 60 mm/m has been found to be 103 T/m².

10.0 SUPPORT CAPACITY ESTIMATION

The capacity of the power support for working of No. 1 seam of Adriyala Block may be calculated from the support resistance after cut, by the following Eq. 6:

$$\text{Support Capacity (in T)} = P_r W_c (L_c + W_d + 0.3) \dots\dots [6]$$

where, P_r is the rated support resistance, T/m²

W_c is the width of the canopy, m

L_c is the length of the canopy, m

W_d is the web depth of shearer, m

The dimensions of the proposed two legged shields with 1150T capacity are $W_c = 1.75\text{m}$, $L_c = 4.79\text{m}$ and $W_d = 0.85\text{ m}$. For the required rated support resistance of 103 T/m^2 , the estimated support capacity is $103 \times 1.75 \times (4.79+0.85+0.577) = 1113\text{ T}$. Thus the proposed two legged shields of 1150T capacity for Adriyala Seam 1 are adequate for given geo mining conditions.

The friable coal immediate roof is more prone to cavity formation at the coal face. Such kind of friable roof geology must be controlled with two legged shield support which will provide active horizontal force towards the coal face due to inclination of the hydraulic legs. This active horizontal force improves the stability of friable roof by preventing slippage along the ubiquitous fracture planes in the coal roof. A two legged shield provides a resultant vertical force nearer to the coal face than four legged shield and this will generate a higher canopy tip load and tip contact. The four legged shield support provide a resultant force away from the coal face and this will result in fractured roof at the rear end of the canopy in friable coal roof conditions as existing in Seam I of Adriyala Project. This will give rise to a tilted canopy and the contact line of the canopy with the coal roof will be far away from the coal face. Such conditions will increase the cavity formation at the face. Thus to prevent formation of cavities ahead of the support and fractured roof conditions two legged shield support is recommended for No. 1 Seam of Adriyala Project.